UNIT 2: SHAPES AND REGIONS

SESSION 1: BINARY SHAPE ANALYSIS

SLO1: EXTRACT AND INTERPRET SHAPE FEATURES FROM BINARY IMAGES FOR OBJECT IDENTIFICATION AND CLASSIFICATION.

Activity 1: Quick Check (3 Minutes)
Fill in the blanks

1. A binary image contains only 0 & 1 pixel values.
2. In a binary image, **1 represents** the object and **0 represents** the background.
3. The number of pixels belonging to an object is called area.
4. The center point of an object is called centroid.

Activity 2: Identify Object Pixels (4 Minutes)
Consider the following binary image:
0 0 0 0 0
0 1 1 1 0
0 1 1 1 0
0 0 0 0 0
Answer the following:

1. Total number of **object pixels (1s)** = 6
2. **Area of the object** = 6 pixels
3. Number of **rows containing object pixels** = 2

Activity 3: Think Fast (3 Minutes)
Tick the correct answer.

1. The smallest rectangle enclosing an object is called:
   - ☐ Centroid
   - ☐ Bounding Box
   - ☐ Perimeter
2. Which feature represents the **length of object boundary**?
   - ☐ Area
   - ☐ Perimeter
   - ☐ Aspect Ratio
3. Binary shape analysis is used in:
   - ☐ Medical imaging
   - ☐ OCR
   - ☐ Industrial inspection
   - ☐ All of the above

Quick Reflection (Optional)

Write **one application of binary shape analysis**: Binary shape analysis is used in medical imaging to identify and analyze the shape and size of objects such as tumors or organs.